

INFORMATION TECHNOLOGY CONCEPTS (391)

—OPEN EVENT—

STATE – 2019

DO NOT WRITE ON TEST BOOKLET

TOTAL POINTS

_____ (*100 points*)

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

No more than sixty (60) minutes testing time

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MULTIPLE CHOICE

Identify the choice that best completes the statement or answers the question. Mark “A” if the statement is true. Mark “B” if the statement is false.

1. Which of the following is virtualization software that installs on a server in a similar manner to a normal operating system, which then runs virtual machines?
 - A. Type 1 hypervisor
 - B. Type 2 hypervisor
 - C. Hardware-assisted virtualization
 - D. Server-side virtualization
2. One benefit of virtualization is that it allows for better use of already existing hardware that may *not* be seeing full utilization.
 - A. True
 - B. False
3. Which type of virtualization client is also called a dumb terminal because it does *not* have an operating system?
 - A. Fat client
 - B. Thin client
 - C. Zero client
 - D. Thick client
4. Virtual machines do *not* require dedicated space and can make direct use of the file system of a host for storing operating system files.
 - A. True
 - B. False
5. What does the term VDI refer to?
 - A. creating VMs on desktop computers
 - B. the use of server-side virtualization to provide virtual desktops
 - C. installing a hypervisor on all client computers
 - D. the installation of host-based virtualization software
6. Logs created by application running Linux are likely to be found in which of the following directories?
 - A. /root
 - B. /lib
 - C. /bin
 - D. /var
7. The current Linux boot loader used to manage a dual boot is LILO.
 - A. True
 - B. False

8. Which Linux command can be used to create a blank file in the current directory?
 - A. vi
 - B. man
 - C. touch
 - D. cat
9. The Mission Control app can be used to manage multiple desktop screens in OSX.
 - A. True
 - B. False
10. Which of the following apps provides tools for customizing the Mac interface?
 - A. Finder
 - B. System Preferences
 - C. Launchpad
 - D. Keychain
11. Communication between a web browser and a web server is an example of which model of communication?
 - A. Client/Server
 - B. Server/Server
 - C. Clone
 - D. Bus
12. An IPv6 address is 128 bits long, and consists of how many blocks of hexadecimal numbers?
 - A. 8
 - B. 32
 - C. 16
 - D. 4
13. What is used to uniquely identify an application on the computer?
 - A. IP Number
 - B. Port Number
 - C. UPC
 - D. ESN
14. An intranet is a private network that uses the TCP/IP protocols.
 - A. True
 - B. False
15. An IP address of 204.162.30.100 belongs in what class of IP addresses?
 - A. Class A
 - B. Class B
 - C. Class C
 - D. Class D

16. What is another name for the Ethernet port that is used to connect devices to a wired network?
 - A. RJ-10
 - B. USB
 - C. Hub
 - D. RJ-45
17. Cellular devices that use GSM or LTE require what to be installed in the device?
 - A. Ethernet
 - B. SIM
 - C. GPS
 - D. ESN
18. Which function of a SOHO router blocks unwanted traffic initiated from the Internet and can restrict Internet access for local computers?
 - A. hostname
 - B. firewall
 - C. wireless access point
 - D. switch
19. Which router feature, when enabled, helps computers on the local network automatically discover and communicate with services provided by other computers on the local network?
 - A. Port forwarding
 - B. Universal Plug and Play (UPnP)
 - C. Port triggering
 - D. Quality of Service (QoS)
20. What type of server can find an IP address for a computer when the fully qualified domain name is known?
 - A. DHCP
 - B. DNS
 - C. PPP
 - D. NetBIOS
21. What term refers to the computer bringing itself up to a working state?
 - A. Booting
 - B. StartOS
 - C. BIOS
 - D. Operating System
22. Which type of boot is also known as a restart?
 - A. Cold boot
 - B. Long boot
 - C. Soft boot
 - D. Restart boot

23. What type of chip does UEFI use to store device drivers and information about Secure Boot?
- A. SDRAM
 - B. SODIMM
 - C. UPC
 - D. NVRAM
24. Which program loads the graphical interface and starts the client/server run-time subsystem during the boot process?
- A. Session manager
 - B. Windows logon
 - C. WinLoad
 - D. Userinit
25. Which of the following programs starts the Service Control Manager?
- A. explorer.exe
 - B. smss.exe
 - C. winlogon.exe
 - D. lsass.exe
26. In Windows 8, the “F8 key at startup” feature is enabled by default.
- A. True
 - B. False
27. You can use a Windows 8 recovery drive on a broken computer that was created on a different computer.
- A. True
 - B. False
28. What is the name given to a lean OS that can be launched to solve Windows startup problems and provides a graphical and command-line interface?
- A. Windows Recovery Environment
 - B. WindowsL
 - C. WindowsLE
 - D. Startup OS
29. Which bootrec command line option is used to repair the boot sector of the system partition?
- A. /scanOS
 - B. /rebuildBCD
 - C. /repairboot
 - D. /fixboot

30. Which of the following boots the OS with a minimum configuration and can be used to solve problems with a new hardware installation or problems caused by user settings?
- A. Windows RE
 - B. Recovery Console
 - C. Safe Mode
 - D. Normal mode
31. Compared with main memory, secondary storage _____.
- A. is faster
 - B. is less expensive
 - C. is more volatile
 - D. has less capacity
32. Which of the following is a multiple-user system?
- A. mainframe computers
 - B. thin client computers
 - C. workstations
 - D. nettop computers
33. _____ is the backbone of a computer, connecting all of its components including the central processing unit (CPU) and primary storage and providing connectors for peripheral devices such as printers, external hard drives, sound cards, and video cards.
- A. The motherboard
 - B. A register
 - C. A bus
 - D. The microprocessor
34. _____ is an advanced optical disc technology that can store up to 3.9 terabytes, roughly 75–150 times more data than the Blu-ray optical disc system although it is the same size and shape as a regular digital video disc (DVD).
- A. The holographic versatile disc
 - B. The motherboard
 - C. A compact disc read-only memory
 - D. A magnetic disc
35. _____ is a type of high-speed memory that a processor can access more rapidly than main memory.
- A. Virtual memory
 - B. Cloud memory
 - C. Volatile memory
 - D. Cache memory

36. Which of the following statements is *true* of organic light-emitting diode (OLED)?
- A. It enables improved contrast than a liquid crystal display (LCD).
 - B. It consumes more power than a light-emitting diode (LED).
 - C. It employs a backlight.
 - D. It produces dull images.
37. The faster the clock speed of the central processing unit (CPU), the ____.
- A. more memory is required for processing
 - B. more heat the processor generates
 - C. larger the chip size required
 - D. slower the machine cycle time
38. The use of middleware to connect disparate systems has evolved into an approach for developing software and systems called ____.
- A. cloud computing
 - B. rapid application development
 - C. service-oriented architecture
 - D. software as a service
39. Which of the following statements is *true* of a database?
- A. It is accessed primarily for use by an organization's database administrator and data administrator.
 - B. It provides an essential foundation for an organization's knowledge management and artificial intelligence systems.
 - C. It is a collection of unstructured data.
 - D. It helps companies analyze information to open new market opportunities.
40. Each attribute in a relational database model can be constrained to a range of allowable values called a ____.
- A. bandwidth
 - B. scope
 - C. slot
 - D. domain
41. Which of the following statements is *true* of radio frequency range for wireless communications?
- A. It operates in the 300 MHz–300 GHz range.
 - B. Its costs are increasing.
 - C. It supports mobile users.
 - D. It is less susceptible to interception.

42. One way to help protect the security of communications over both wired and wireless networks is to _____.
A. encrypt the data
B. use radio waves
C. enable infrared transmission
D. keep using the same password
43. A _____ is a web page that combines useful information and links and acts as an entry point to the web.
A. web browser
B. web design framework
C. web log
D. web portal
44. _____ is *not* a web programming language.
A. Java
B. JavaScript
C. Hypertext Preprocessor
D. COBOL
45. _____ are specially denoted text or graphics on a web page that, when clicked, open a web page containing related content.
A. Cookies
B. Tags
C. Hyperlinks
D. Registers
46. Video content can be placed on the Internet using the same approach as a blog.
A. True
B. False
47. Each computer on the Internet has an assigned address, called its Internet Protocol (IP) address, that identifies it on the Internet.
A. True
B. False
48. The United States has high Internet penetration among its population constituting the majority of people online.
A. True
B. False

49. Which of the following statements is *true* of retargeting?
- A. It is targeted at approximately 50% of website visitors who shop using smartphone applications.
 - B. It employs personalized ads to direct shoppers back to a retailer's website they previously abandoned.
 - C. It attempts to change buying habits of web shoppers to get them to consider products they have never considered purchasing before.
 - D. It attempts to change buying habits of web shoppers by getting them to visit websites they have not visited before.
50. Authentication technologies are used by many organizations to _____.
- A. verify the accuracy of their product descriptions and prices
 - B. secure sensitive data during e-commerce
 - C. confirm the uniform resource locator (URL) of their suppliers' websites
 - D. confirm the identity of a user requesting access to information or assets